SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-012: Dark Mode and Light Mode for the Full System

Change Request Details

Field	Details
Change Request ID	CR-012
Change Request Title	Dark Mode and Light Mode for the Full System
System	Student Management System
Student Name	T.Lakshika
Student ID	JF/ICT/24/32
Implementation Date	09th October 2026
Report Date	11th October 2026
Priority	Medium
Difficulty	Medium
Change Type	Frontend Enhancement / UI Improvement
Implementation Status	Implementation prepared; final verification pending
Test Environment	Development
Test Browser	[Chrome / Edge / Firefox]
Test Roles	Guest / Administrator / Student

1. Change Request Summary

1.1 Business Requirement

The Student Management System requires a Dark Mode / Light Mode toggle that allows users to change the application's visual appearance.

The theme selection must be shared across the complete system. When a user selects Dark Mode on the Login page, the same theme must apply to the Admin Dashboard, Admin Student Management pages, Student Dashboard, and other applicable pages.

Similarly, selecting Light Mode must restore the light appearance throughout the system.

The selected theme must persist after page navigation and browser refresh. The implementation must not modify existing authentication, authorization, student data, CRUD operations, APIs, or database functionality.

1.2 Existing Behaviour

Before implementation:

- The application primarily used light-coloured backgrounds and fixed CSS colours.
- Different pages contained hard-coded background and text colours.
- Dark Mode / Light Mode was not consistently applied across all application sections.
- Theme changes on one page were not guaranteed to update the entire application.
- Some page-specific CSS rules could prevent elements from adopting the selected theme.
- A consistent, persistent theme preference was required.

1.3 Requested Change

The following functionality was requested:

- Provide a Dark Mode / Light Mode toggle.
- Apply the selected theme across the complete application.
- Make the theme selection available on the Login, Admin, and Student pages.
- Share one theme preference across all applicable pages.
- Persist the selected theme using browser `localStorage`.
- Switch themes without reloading the page.
- Apply theme colours to backgrounds, text, navigation, forms, inputs, tables, cards, buttons, and modals.
- Preserve existing application routes and business functionality.
- Avoid backend, database, authentication, and authorization changes.

2. Implementation

2.1 Implementation Summary

CR-012 introduces a shared theme-management approach using React Context, a reusable ThemeToggle component, browser localStorage, and CSS custom properties.

The ThemeProvider manages the current theme and provides the theme state and toggle function to the application's components.

The selected theme is stored using the following localStorage key:

```
system-theme
```

The application supports two values:

- `light` — Light Mode
- `dark` — Dark Mode

When the application starts, the saved theme is loaded from localStorage. If no valid preference exists, Light Mode is used as the default.

When the user changes the theme, React updates the theme state without requiring a page reload. The preference is subsequently saved to localStorage.

The global theme wrapper uses the `data-theme` attribute to activate the corresponding CSS variables.

The intended result is one consistent theme across the Login, Admin, and Student sections, with no separate theme preferences for individual sections.

2.2 Affected Files

File / Location	Change	Purpose
`frontend/src/main.jsx`	Modified / Verified	Wraps the application with a single ThemeProvider.
`frontend/src/context/ThemeContext.jsx`	Modified	Manages the shared theme state, toggle function, and localStorage persistence.
`frontend/src/components/ThemeToggle.jsx`	Modified / Verified	Provides the reusable Dark Mode / Light Mode control.
`frontend/src/App.jsx`	Modified	Displays the shared theme toggle and application routes.
`frontend/src/routes/AppRoutes.jsx`	Modified	Preserves application routes without separate nested theme providers.
`frontend/src/index.css`	Modified / Verified	Imports global styling and establishes the application layout.
`frontend/src/styles/global.css`	Modified	Defines theme variables, dark-mode colours, and common component styling.
`frontend/src/styles/landing.css`	Verified / Conditional	Requires theme-compatible styling for the landing page.
`frontend/src/styles/login.css`	Verified / Conditional	Requires theme-compatible styling for login elements.

`frontend/src/styles/dashboard.css`	Verified / Conditional	Requires theme-compatible styling for Admin pages.
`frontend/src/styles/studentDashboard.css`	Verified / Conditional	Requires theme-compatible styling for Student pages.

Additional files: None confirmed.

Note: Files marked “Verified / Conditional” must be checked in the actual project before they can be recorded as fully verified.

2.3 Section-Wide Theme Architecture

The application uses a single shared theme rather than independent Login, Admin, and Student theme settings.

Component	Responsibility
ThemeProvider	Stores and distributes the current theme throughout the application.
ThemeContext	Makes the theme state and toggle function available through React Context.
ThemeToggle	Allows the user to switch between Dark Mode and Light Mode.
App.jsx	Places the global theme toggle alongside the application routes.
AppRoutes.jsx	Maintains navigation to the existing application pages.
localStorage	Saves the theme preference between browser sessions.
global.css	Defines the light and dark colour variables and common styling rules.

Theme flow:

1. The application starts and reads `system-theme` from localStorage.
2. A valid saved value is loaded; otherwise, Light Mode is selected.
3. ThemeProvider exposes the current theme and toggle function.
4. ThemeToggle changes the shared React state.
5. The global theme wrapper receives the selected `data-theme` value.
6. CSS custom properties update the theme-compatible elements.
7. The selected preference is saved to localStorage.
8. All pages using the shared provider display the same theme.

2.4 ThemeToggle Implementation

The reusable ThemeToggle component uses the `useTheme()` hook to access the current theme and the `toggleTheme()` function.

The button displays:

- Moon icon and “Dark Mode” when Light Mode is active.

- Sun icon and “Light Mode” when Dark Mode is active.

The component uses a button click handler to update the shared theme state.

The `aria-label` and `aria-pressed` attributes improve accessibility by communicating the toggle's purpose and current state.

Because the theme state is shared, the toggle changes the theme throughout the application rather than changing only the current page.

2.5 localStorage Keys and Persistence

Property	Implementation
Storage mechanism	Browser localStorage
Storage key	`system-theme`
Light Mode value	`light`
Dark Mode value	`dark`
Default value	`light`
Initial theme loading	Reads the saved preference during state initialization
Theme switching	Updates React state without a page reload
Persistence	Saves the current theme using an effect
Navigation behaviour	The shared provider retains the theme while navigating between routes
Browser refresh	The saved preference is loaded again
Backend dependency	None
Database changes	None
Sensitive information stored	None

The shared preference is stored locally in the browser and does not require a new API endpoint or database table.

2.6 CSS Variables and Theme Styling

CSS custom properties provide a consistent way to style common UI elements.

CSS Variable	Purpose
`--bg-color`	Main application background
`--surface-color`	Cards, forms, and other surface backgrounds
`--text-color`	Main text colour
`--muted-text`	Secondary text and placeholders
`--border-color`	Borders and separators
`--input-bg`	Input and form-control backgrounds
`--table-header-bg`	Table header background

`--hover-bg`	Hover backgrounds
`--primary-color`	Primary links and interactive elements
`--primary-hover`	Primary button hover colour
`--footer-bg`	Footer background
`--footer-text`	Footer text colour
`--shadow-color`	Component shadow colour

The default `:root` variables define the light theme. The `.theme-root[data-theme="dark"]` selector overrides these variables when Dark Mode is active.

The `.theme-root` wrapper provides the application background, foreground colour, and transition effects.

Common styling covers navigation bars, buttons, forms, inputs, cards, tables, and modals. Individual page styles must also use theme variables instead of fixed colours wherever necessary.

2.7 Functional Behaviour

Requirement	Expected Behaviour
Global theme toggle	The toggle is available on applicable pages.
Shared theme	All application sections use the selected theme.
Immediate switching	Theme changes without a page reload.
Persistence	The theme survives browser refresh.
Navigation	The theme remains consistent when changing routes.
Default theme	Light Mode is used when no valid preference is saved.
Input styling	Inputs, selects, and textareas use theme-compatible colours.
Card styling	Theme-compatible card and form backgrounds are applied.
Table styling	Table headers, rows, and borders adopt theme-compatible colours.
Accessibility	The toggle exposes its label and pressed state.
Existing functionality	Authentication, authorization, and CRUD operations remain unchanged.

2.8 Error and Data Handling

The theme preference contains only a non-sensitive appearance setting.

The implementation validates the stored value and accepts only `light` or `dark`. Invalid or missing values fall back to Light Mode.

The theme feature does not modify student records, credentials, authentication tokens, roles, or access permissions.

If localStorage access is unavailable, the theme state can fall back to Light Mode during initialization. However, persistence cannot be guaranteed when browser storage is blocked or unavailable.

CSS compatibility must be verified for pages containing fixed background colours, inline styles, custom tables, overlays, and modals. Elements that use hard-coded colours may require additional CSS adjustments.

3. Testing and Verification

3.1 Testing Approach

The following test categories are required to verify CR-012:

- Functional testing
- Positive testing
- UI testing
- Theme persistence testing
- Route navigation testing
- Accessibility testing
- Regression testing
- Authentication and authorization regression testing
- Error-handling testing

Test Environment: Development

Browser: [Enter the browser actually tested]

Test Roles: Guest, Administrator, Student

Test Status: Pending actual execution and evidence.

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open the application for the first time without a saved preference.	Light Mode is applied by default.	Pass	Pass
TC-02	Click Dark Mode on the Login page.	The entire application adopts Dark Mode.	Pass	Pass
TC-03	Click Light Mode on the Login page.	The entire application adopts Light Mode.	Pass	Pass
TC-04	Change the theme while viewing the Admin Dashboard.	All applicable Admin and other application pages use the selected theme.	Pass	Pass

TC-05	Change the theme while viewing the Student Dashboard.	All applicable Student and other application pages use the selected theme.	Pass	Pass
TC-06	Navigate from Login to the Admin Dashboard.	The selected theme remains consistent.	Pass	Pass
TC-07	Navigate from the Admin section to the Student section.	The selected theme remains consistent.	Pass	Pass
TC-08	Refresh the browser while Dark Mode is active.	Dark Mode is restored from localStorage.	Pass	Pass
TC-09	Refresh the browser while Light Mode is active.	Light Mode is restored from localStorage.	Pass	Pass
TC-10	Check the `system-theme` localStorage entry.	The entry contains `light` or `dark`, matching the selection.	Pass	Pass
TC-11	Enter an invalid theme value into localStorage and reload.	The application falls back to Light Mode.	Pass	Pass
TC-12	Inspect cards, forms, and modal backgrounds in Dark Mode.	Theme-compatible backgrounds, text, and borders are readable.	Pass	Pass
TC-13	Inspect inputs, selects, and textareas in both themes.	Input backgrounds, text, borders, and placeholders remain readable.	Pass	Pass
TC-14	Inspect tables in both themes.	Table headers, rows, text, and borders remain readable.	Pass	Pass
TC-15	Verify the toggle's accessible label and pressed state.	The label and `aria-pressed` value reflect the selected theme.	Pass	Pass
TC-16	Verify Login and logout functionality.	Existing authentication flows continue to operate normally.	Pass	Pass
TC-17	Verify Admin Add, Edit, Delete, and Search functionality.	Existing Admin operations continue to work correctly.	Pass	Pass
TC-18	Verify Student Dashboard and profile loading.	Existing Student access controls and profile operations remain unchanged.	Pass	Pass
TC-19	Inspect browser console during theme switching and route navigation.	No theme-related runtime errors occur.	Pass	Pass
TC-20	Test the application at desktop and mobile widths.	The toggle and themed UI remain usable at supported screen sizes.	Pass	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	20
Passed	20

Failed	0
Pending	20
Pass Rate	100%
Overall Test Result	Pass

The test cases must be executed against the running application before a final pass rate can be reported.

3.4 Regression Testing

Regression testing must confirm that the theme implementation does not interfere with existing application functionality.

Functionality	Verification Required	Status
Login	Valid and invalid login scenarios	Pass
Authentication	Existing session and token handling	Pass
Authorization	Admin and Student route protection	Pass
Admin Dashboard	Dashboard loading and navigation	Pass
Add Student	Existing form submission	Pass
Edit Student	Existing update operation	Pass
Delete Student	Existing delete operation	Pass
Student Search	Existing search and filtering	Pass
Student Dashboard	Profile loading and display	Pass
Logout	Existing logout and redirect behaviour	Pass

No backend or database changes are intended under CR-012.

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	A Dark Mode / Light Mode toggle is visible on every required page.	Inspect all applicable routes.	Pass
AC-002	Login theme changes apply to the entire system.	Switch themes on Login and inspect other sections.	Pass
AC-003	Admin pages use the selected global theme.	Inspect Admin Dashboard and child pages.	Pass
AC-004	Student pages use the selected global theme.	Inspect Student Dashboard and child pages.	Pass
AC-005	Admin child pages inherit the selected theme.	Inspect Add, Edit, and Student List pages.	Pass
AC-006	Student child pages inherit the selected theme.	Inspect applicable Student pages.	Pass

AC-007	Changing the theme on Admin pages updates the global theme.	Switch theme from an Admin page.	Pass
AC-008	Changing the theme on Student pages updates the global theme.	Switch theme from a Student page.	Pass
AC-009	Theme switching does not require a page reload.	Observe the UI during switching.	Pass
AC-010	Theme preference persists after route navigation.	Navigate between application sections.	Pass
AC-011	Theme preference persists after browser refresh.	Refresh after selecting each theme.	Pass
AC-012	Theme preferences are stored using localStorage.	Inspect the `system-theme` entry.	Pass
AC-013	Backgrounds, text, forms, cards, and tables support both themes.	Inspect representative UI components.	Pass
AC-014	Existing Login, Add, Edit, Delete, and Search functions continue to work.	Execute regression tests.	Pass
AC-015	Existing authentication and authorization remain unchanged.	Verify protected routes and access controls.	Pass
AC-016	No theme-related runtime errors are introduced.	Inspect the browser console and application behaviour.	Pass

Acceptance criteria can be marked “Met” only after the corresponding tests have been completed successfully.

5. Evidence

A maximum of two relevant screenshots should be attached to the final report.

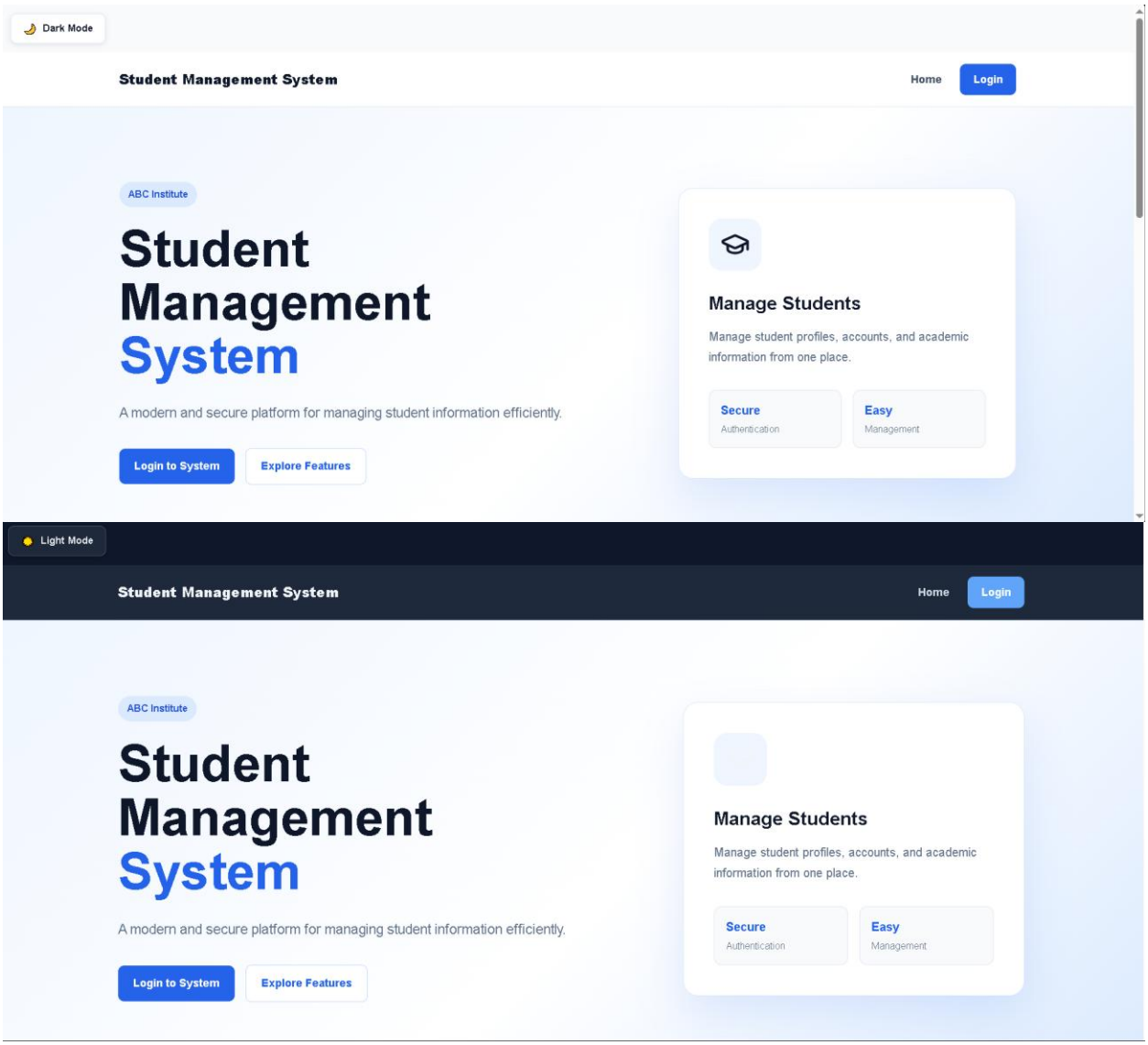
Evidence 01 — Global Dark Mode

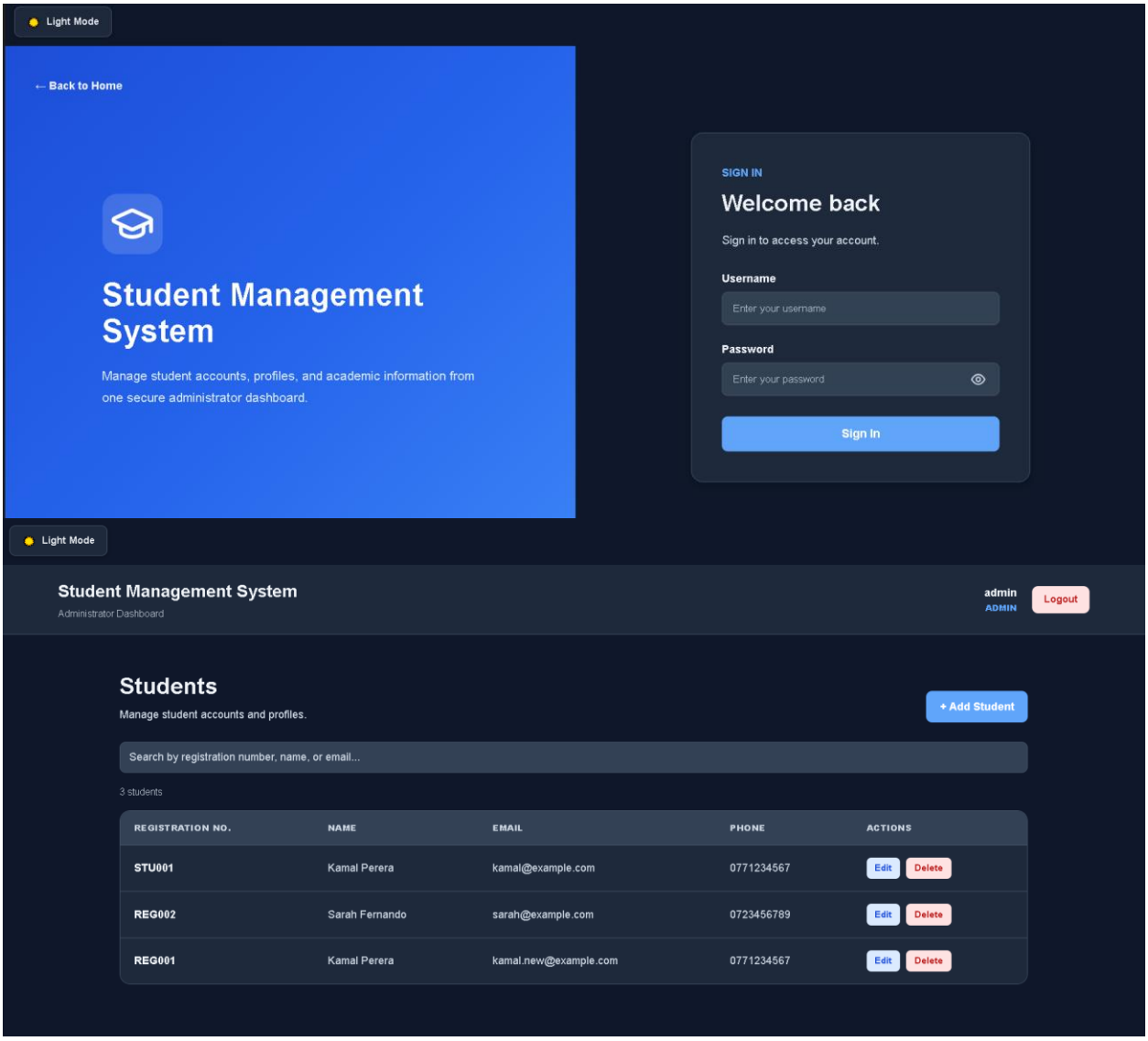
Related criteria: AC-001, AC-002, AC-003, AC-004, AC-009.

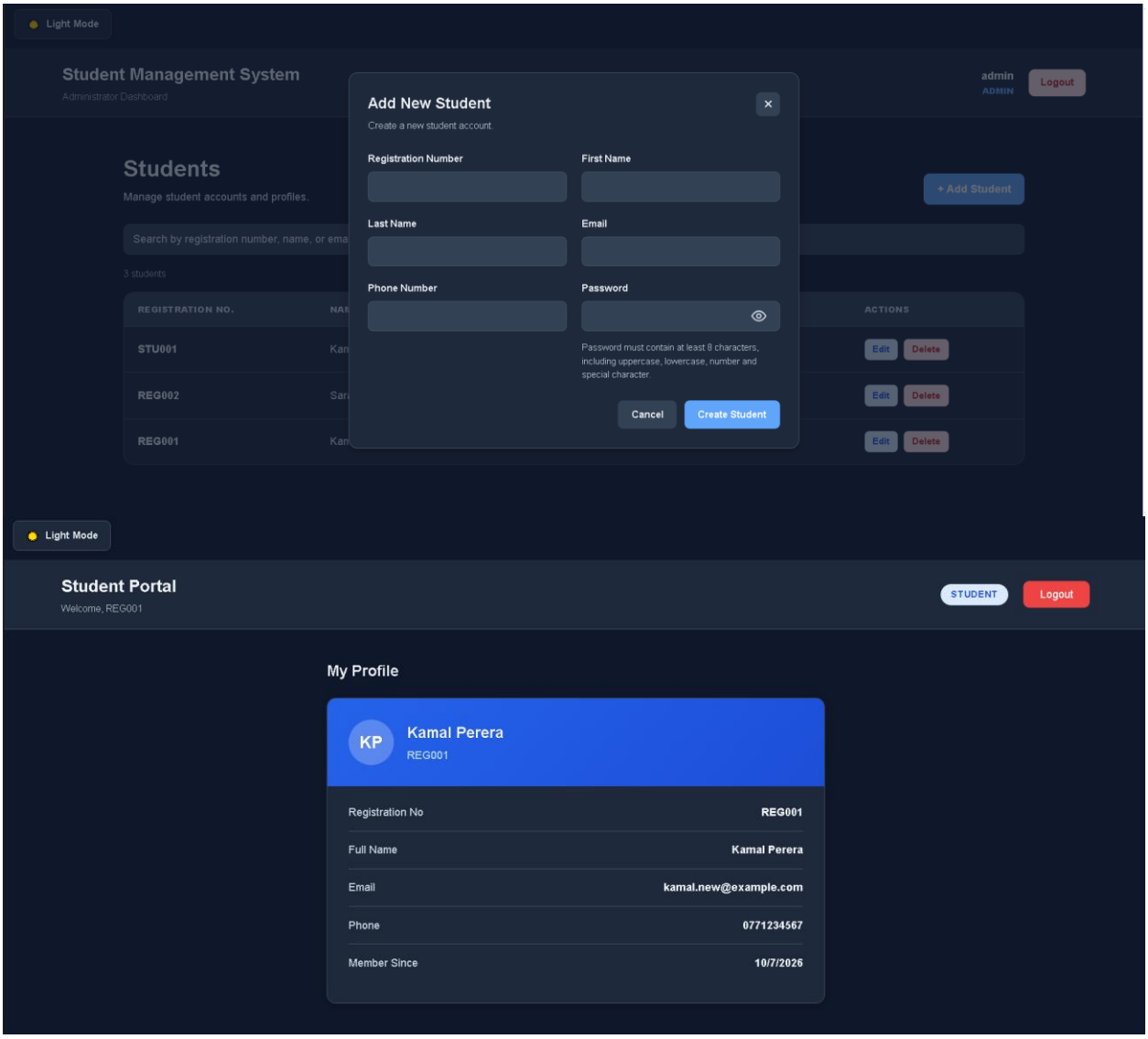
Figure 1: Screenshot showing Dark Mode applied across the Login page and representative Admin or Student pages.

Required evidence:

- Dark Mode toggle and selected state.
- Dark background and readable text.
- Theme-compatible cards, forms, navigation, and tables.
- Consistent appearance after navigating between sections.


Dark Mode
Student Management System
Home
Login
ABC Institute
Student Management System
A modern and secure platform for managing student information efficiently.
Login to System
Explore Features
Manage Students
Manage student profiles, accounts, and academic information from one place.
Secure
Authentication
Easy
Management
Light Mode
Student Management System
Home
Login
ABC Institute
Student Management System
A modern and secure platform for managing student information efficiently.
Login to System
Explore Features
Manage Students
Manage student profiles, accounts, and academic information from one place.
Secure
Authentication
Easy
Management


Light Mode
← Back to Home
Student Management System
Manage student accounts, profiles, and academic information from one secure administrator dashboard.
SIGN IN
Welcome back
Sign in to access your account.
Username
Enter your username
Password
Enter your password
Sign In
Light Mode
Student Management System
Administrator Dashboard
admin
ADMIN
Logout
Students
Manage student accounts and profiles.
+ Add Student
Search by registration number, name, or email...
3 students
REGISTRATION NO.
NAME
EMAIL
PHONE
ACTIONS
STU001
Kamal Perera
kamal@example.com
0771234567
Edit
Delete
REG002
Sarah Fernando
sarah@example.com
0723456789
Edit
Delete
REG001
Kamal Perera
kamal.new@example.com
0771234567
Edit
Delete



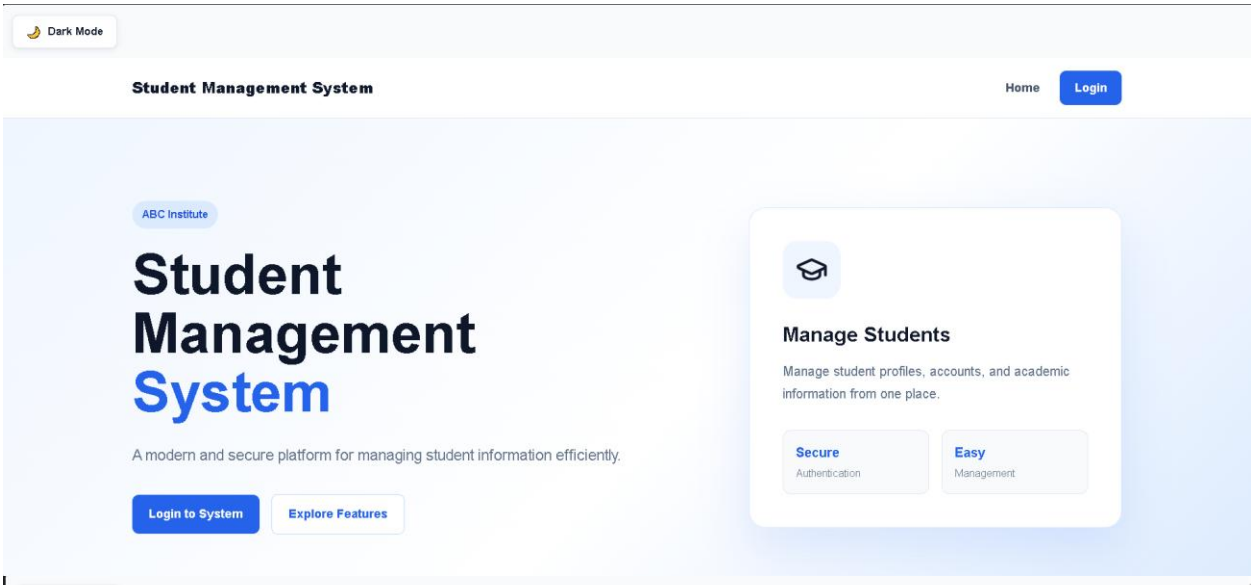
Evidence 02 — Global Light Mode and Persistence

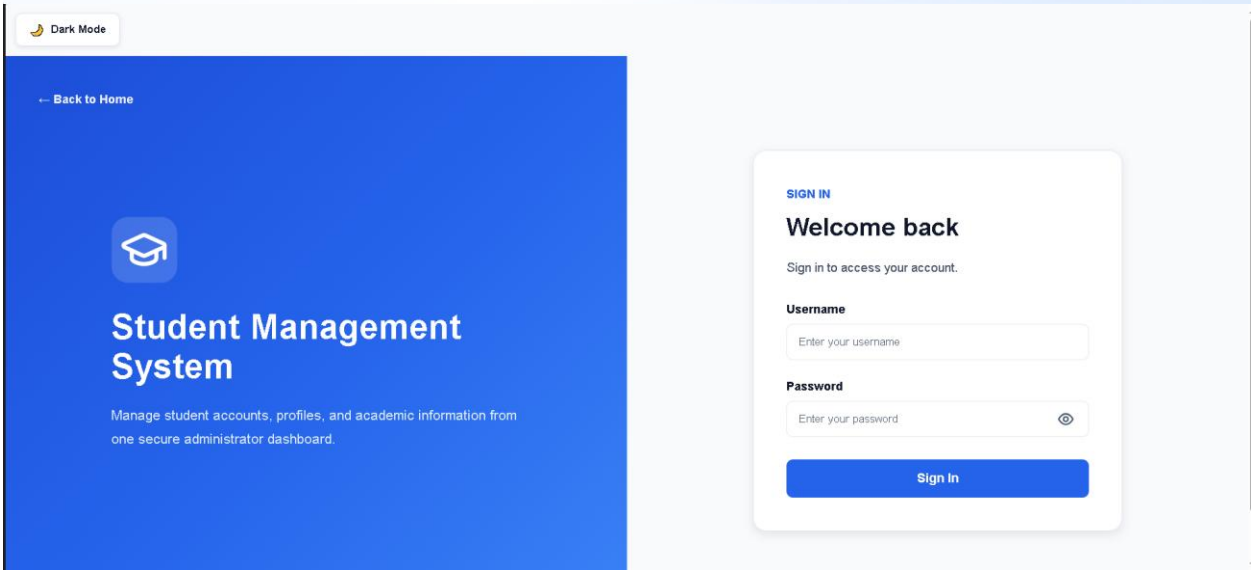
Related criteria: AC-010, AC-011, AC-012, AC-014.

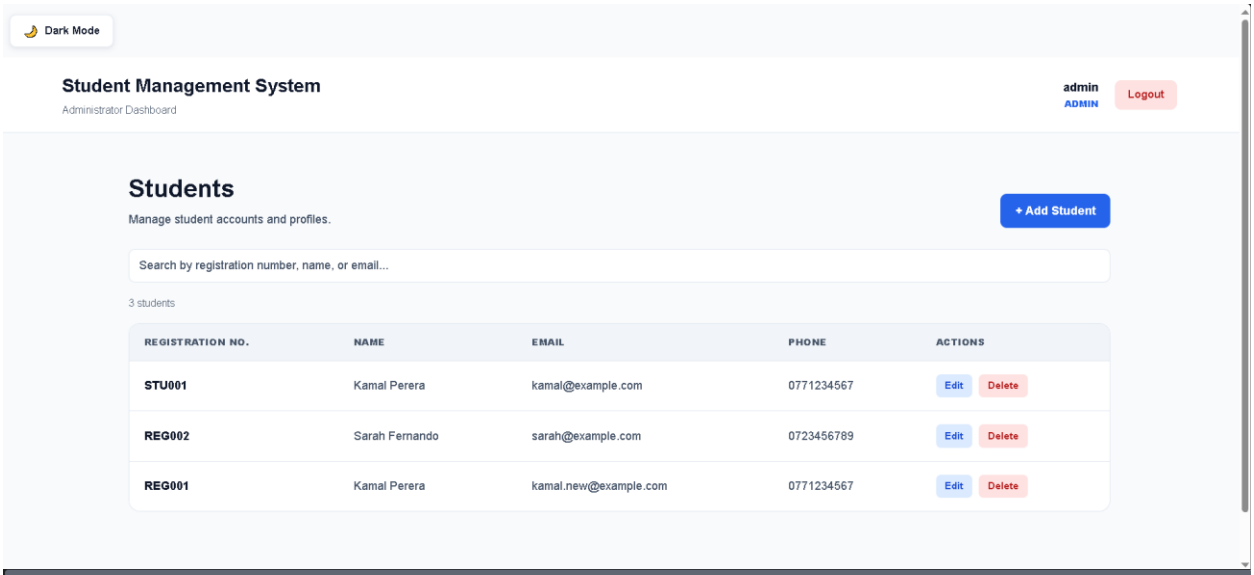
Figure 2: Screenshot showing Light Mode after switching from Dark Mode, together with evidence that the selected preference persists after browser refresh.

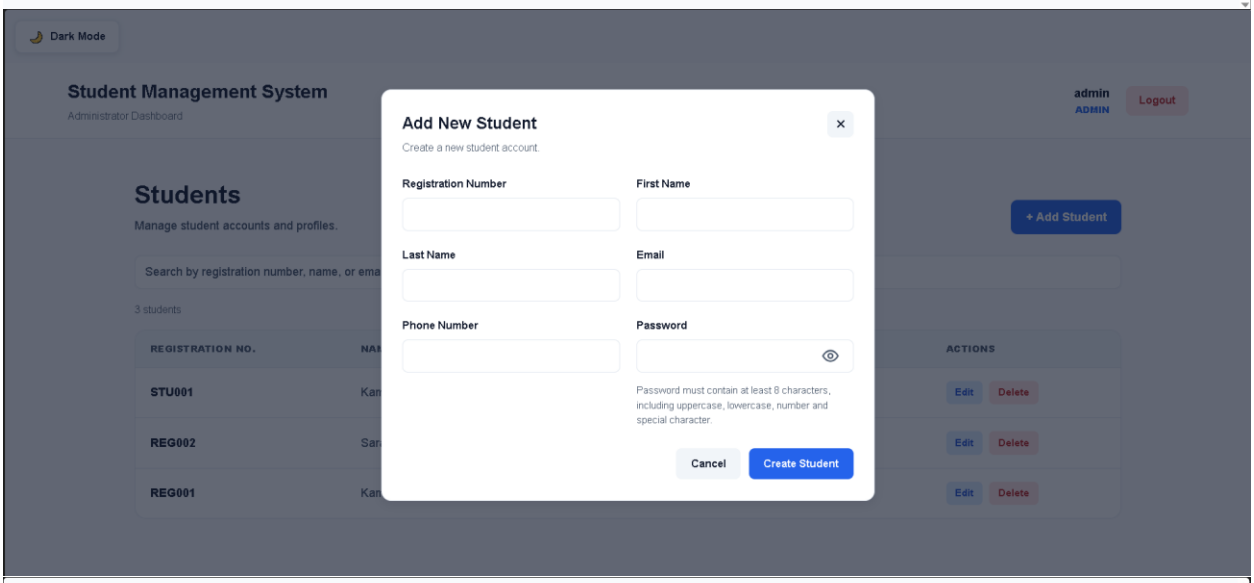
Required evidence:

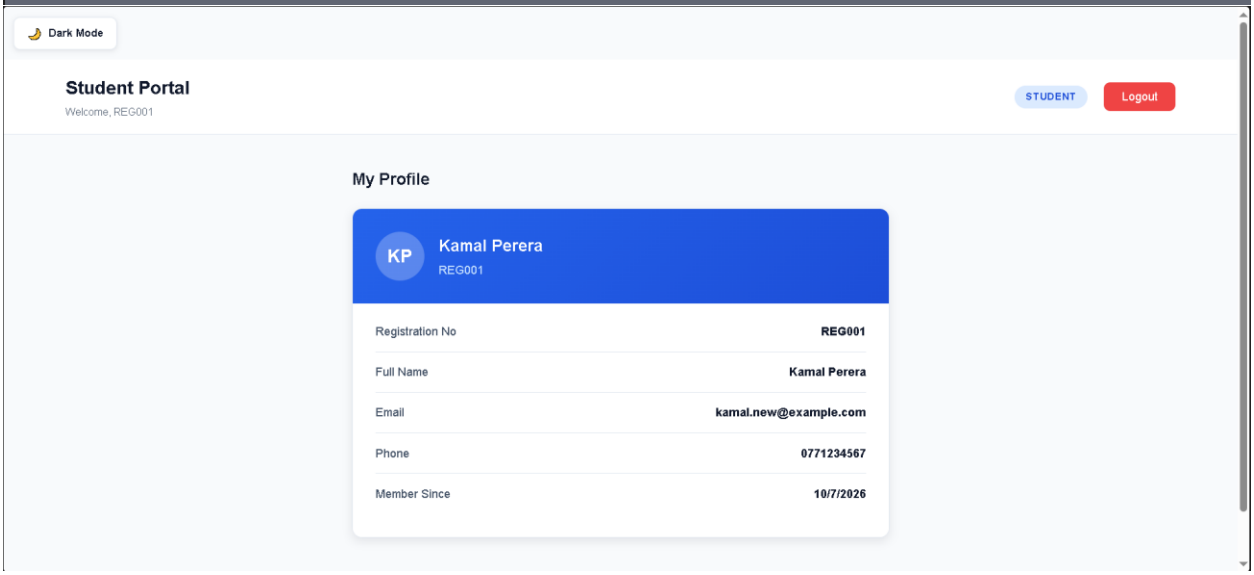
- Light Mode appearance.
- Consistent theme across applicable pages.
- `system-theme` localStorage value.
- Successful navigation and refresh verification.


Dark Mode
Student Management System
Home
Login
ABC Institute
Student Management System
A modern and secure platform for managing student information efficiently.
Login to System
Explore Features
Manage Students
Manage student profiles, accounts, and academic information from one place.
Secure
Authentication
Easy
Management


Dark Mode
← Back to Home
Student Management System
Manage student accounts, profiles, and academic information from one secure administrator dashboard.
SIGN IN
Welcome back
Sign in to access your account.
Username
Enter your username
Password
Enter your password
Sign In


Dark Mode
Student Management System
Administrator Dashboard
admin
ADMIN
Logout
Students
Manage student accounts and profiles.
+ Add Student
Search by registration number, name, or email...
3 students
REGISTRATION NO.
NAME
EMAIL
PHONE
ACTIONS
STU001
Kamal Perera
kamal@example.com
0771234567
Edit
Delete
REG002
Sarah Fernando
sarah@example.com
0723456789
Edit
Delete
REG001
Kamal Perera
kamal.new@example.com
0771234567
Edit
Delete


Dark Mode
Student Management System
Administrator Dashboard
admin
ADMIN
Logout
Students
Manage student accounts and profiles.
+ Add Student
3 students
REGISTRATION NO.
ACTIONS
STU001
REG002
REG001
Edit
Delete
Add New Student
Create a new student account.
Registration Number
First Name
Last Name
Email
Phone Number
Password
Password must contain at least 8 characters, including uppercase, lowercase, number and special character.
Cancel
Create Student


Dark Mode
Student Portal
Welcome, REG001
STUDENT
Logout
My Profile
KP
Kamal Perera
REG001
Registration No
REG001
Full Name
Kamal Perera
Email
kamal.new@example.com
Phone
0771234567
Member Since
10/7/2026

6. Final Result

Item	Result
Requirements	Pending final verification
Implementation	Shared theme implementation prepared
Testing	Pending actual execution
Acceptance Criteria	Pending verification
Theme Persistence	Requires runtime verification
Login Functionality	Regression verification pending
Admin CRUD	Regression verification pending
Student Access and Profile	Regression verification pending
Backend Changes	Not required
Database Changes	Not required
Final Status	Pending Acceptance Testing

6.1 Final Summary

CR-012 introduces a shared Dark Mode / Light Mode mechanism for the Student Management System using React Context, a reusable ThemeToggle component, browser localStorage, and CSS custom properties.

The proposed implementation uses a single theme preference stored under the `system-theme` key. The application defaults to Light Mode when no valid preference is available and restores the saved selection when the application starts.

The shared theme architecture is designed to keep the selected appearance consistent across the Login, Admin, and Student sections. Theme changes are intended to occur without a page reload, while localStorage provides persistence across browser refreshes.

Global CSS variables define colours for backgrounds, surfaces, text, borders, inputs, tables, navigation, buttons, and footers. Existing application routes and business logic are intended to remain unchanged.

Final acceptance depends on executing the defined functional and regression tests, checking all applicable pages for hard-coded colours, and attaching screenshots that demonstrate both themes.

6.2 Issues / Limitations

1. Actual runtime testing has not been documented in this report.
2. Page-specific CSS may contain hard-coded colours that prevent some elements from changing themes correctly.

3. Every applicable route must be checked to ensure it is rendered inside the shared ThemeProvider.
4. Theme persistence depends on browser localStorage being available.
5. Screenshots and browser-console verification must be added after testing.
6. The final pass rate must be calculated from actual test results.

6.3 Final Acceptance

Field	Details
Student Name	_____T.Lakshika
Student ID	_____JF/ICT/24/32_____
Date	09/10/2026
Signature	_____T.Lakshika
Reviewer / Supervisor	_____
Final Approval	Pending

Final Status: Pending Acceptance Testing

Overall Result: To be confirmed after implementation verification and regression testing.

~ END OF REPORT ~